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Research on the optimization of integrated energy systems in data centers considering flexible resources

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ABSTRACT

Data centers are becoming one of the most energy-intensive infrastructures, facing increasing pressure from rising electricity costs and stringent carbon reduction targets. To enhance operational flexibility and energy efficiency, this study proposes a coordinated optimization framework for data center integrated energy systems that simultaneously exploits supply-side and demand-side flexible resources. The framework integrates an integrated energy system architecture with computational load shifting and air thermal inertia, enabling coordinated scheduling of electricity, cooling, and heating subsystems under time-varying electricity prices. A mixed-integer linear programming model is developed to minimize operating cost while maintaining acceptable quality of service, with explicit consideration of cascaded waste heat recovery and flexible dispatch of conventional backup resources. A case study based on a large-scale data center in Hebei Province, China, demonstrates that the proposed strategy can reduce annual operating costs by up to 14.7% and achieve approximately 8% carbon emission reduction compared with conventional operation. The results highlight the practical potential of transforming data centers from rigid energy consumers into actively dispatchable entities within integrated energy systems, providing insights for low-carbon and cost-effective data center operation.

Nomenclature

Abbreviation	Full Name
IES	Integrated Energy System
CCHP	Combined Cooling, Heating and Power
AC	Absorption Chiller
QoS	Quality of Service
BES	Battery Energy Storage
MILP	Mixed-Integer Linear Programming
SOC	State of Charge
HTST	Heat Storage Tank
CSR	Cost Savings Rate
ERR	Emission Reduction Rate
PBP	Payback Period

Superscript	Meaning
max	Maximum Value
min	Minimum Value
S/M/L	Task Type-related
DL	Delay Level-related
PT	Task Processing Duration-related
Subscript	**Meaning**
opt	Optimized
trad	Traditional Mode
d	Diesel Generator-related
c	Chiller Unit-related
ac	Absorption Chiller-related

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PV	Photovoltaic
COP	Coefficient of Performance
LHV	Lower Heating Value
Symbol	**Meaning**
Q	Heat/Cooling
E	Electricity/Energy Consumption
P	Power
T	Temperature
η	Efficiency
m	Mass/Quantity
N	Quantity
C	Capacity

hrp	Heat Recovery Pump-related
ct	Cooling Tower-related
DC	Data Center
hx	Heat Exchanger-related
IT	IT Equipment-related
ch	Charging-related
ws	Waste Heat-related
in	Input/Charging-related
out	Discharging-related
batch	Batch Processing Task-related
grid	Grid-related
fuel	Fuel-related

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β	Cooling Tower Energy Consumption Coefficient	heat	Heating-related
X	Load Shifting Coefficien	t	Time-related
r	Standard Temperature Coefficient of Photovoltaic Panels		

1. Introduction

Data centers have become one of the most electricity-intensive infrastructures in modern energy systems, driven by the rapid expansion of cloud computing, artificial intelligence, and large-scale data services. The continuous growth in computational demand has led to a sharp increase in electricity consumption, operating costs, and associated carbon emissions. In China, data centers consumed approximately 150 TWh of electricity in 2023 [1], accounting for about 1.6% of national electricity demand, and this figure is projected to exceed 500 TWh by 2030 [2]. Under increasingly stringent carbon mitigation targets and rising energy prices, improving the energy efficiency and operational flexibility of data centers has become a critical challenge for both energy systems and digital infrastructure planning [3,4].

The rapid development of distributed energy technologies and the advancement of the energy internet concept have laid a solid technical foundation for achieving multi-energy complementarity and flexible dispatch in data centers [5–7]. As a key enabler of the energy internet, integrated energy systems (IES) realize the complementary utilization and optimal allocation of multiple energycarriers—electricity, heat, and gas—through combined cooling, heating, and power (CCHP) technologies together with the coordinated integration of renewable energy sources such as photovoltaic and wind power, thereby significantly enhancing overall system efficiency and cleanliness [8–10]. Within this framework, integrating data centers into IES can not only reduce energy supply costs and environmental impacts but also improve operational flexibility and system security, providing a viable pathway toward efficient and sustainable data center operation [11].

Within the IES framework, data centers are increasingly recognized as high-quality flexible loads due to their adjustable operating time scales and service quality constraints [12]. Existing research has investigated coordinated operation models of electricity–heat systems [13], the utilization of spatiotemporal flexibility of data centers combined with battery energy storage systems to optimize

distribution network operation [14], the impact of data center siting on power system efficiency, as well as energy and exergy analyses of combined heat and power systems [15]. However, most of these studies primarily focus on limited energy dimensions, such as electricity–heat coordination, while the comprehensive representation of data center physical flexibility—particularly the coupling of thermal inertia, workload migration, and waste heat utilization—remains insufficient.

Existing studies have investigated data-center-integrated energy systems from various perspectives, including coordinated electricity–heat operation, renewable energy integration, waste heat recovery, and flexible demand response. Existing research can generally be classified into three categories. First, several studies focus on the coordinated optimization of integrated energy systems and data centers, aiming to improve energy efficiency through multi-energy coupling and waste heat utilization. Second, a growing body of literature explores computational flexibility, such as workload shifting and geographical load migration, to reduce operating costs and support power system balancing. Third, recent studies have begun to exploit thermal flexibility by utilizing the thermal inertia of buildings or server rooms to provide short-term cooling load regulation.

Despite these advances, several important limitations remain. Most existing studies investigate computational load flexibility and thermal inertia separately, without considering their coordinated interaction within a unified optimization framework. Furthermore, waste heat recovery is often modeled in a simplified manner, while the cascading utilization of recovered heat across cooling and heating subsystems is rarely considered. In addition, conventional backup facilities in data centers, such as diesel generators and thermal storage devices, are typically treated as emergency resources rather than actively dispatchable assets. Consequently, the full flexibility potential of data centers within integrated energy systems has not yet been fully exploited.

To address these research gaps, this paper proposes a coordinated optimization framework for a data center integrated energy system considering both computational load shifting and air thermal inertia. By jointly exploiting supply-side and demand-side flexibility resources, the proposed framework enables deeper coupling among electricity, cooling, and heating subsystems, thereby improving operational economy, energy utilization efficiency, and carbon reduction performance.

The main contributions of this study are summarized as follows.

(1) A unified optimization framework integrating computational load shifting and air thermal inertia is developed. Unlike existing studies that investigate these flexibility resources independently, the proposed framework captures their coordinated effects on electricity consumption and cooling demand, enabling a more comprehensive representation of data center flexibility.
(2) A cascaded waste heat utilization mechanism is established by explicitly modeling waste heat recovery, heat upgrading, thermal storage, and absorption cooling processes. This approach strengthens multi-energy coupling and improves overall energy utilization efficiency.
(3) Conventional backup facilities, including diesel generators, battery energy storage systems, and thermal storage tanks, are transformed into actively dispatchable resources under normal operating conditions. The coordinated scheduling of these resources further enhances system operational flexibility and economic performance.

Overall, this study extends existing IES modeling approaches by fully exploiting the multi-dimensional flexibility of data centers and provides a practical pathway toward low-carbon, flexible, and energy-efficient operation. The proposed framework offers valuable insights for the planning and operation of future data centers within integrated energy systems.

2. System modeling

2.1. Integrated energy system

This study considers a data center integrated energy system (DC–IES) that jointly supplies electricity, cooling, and heating to meet both computational and thermal demands. The system architecture is designed to exploit the inherent flexibility of data centers through coordinated operation of energy conversion units, storage facilities, and flexible loads. As illustrated in Fig. 1, the DC–IES consists of grid connection, on-site generation units, cooling and heat recovery subsystems, and multiple forms of energy storage.

Electric power can be supplied by the utility grid and on-site diesel generators. Although diesel generators are traditionally deployed as emergency backup units, this study allows their limited dispatch under normal operating conditions to enhance system flexibility and reduce peak electricity procurement costs. Cooling demand of the data center is primarily satisfied by electric chillers, while waste heat from the cooling process is recovered through heat recovery pumps and further utilized by absorption chillers or stored in thermal storage tanks. This cascaded utilization of waste heat establishes a strong coupling between electricity, cooling, and heating subsystems.

To accommodate the dynamic characteristics of data center operation, battery energy storage systems and thermal storage tanks are

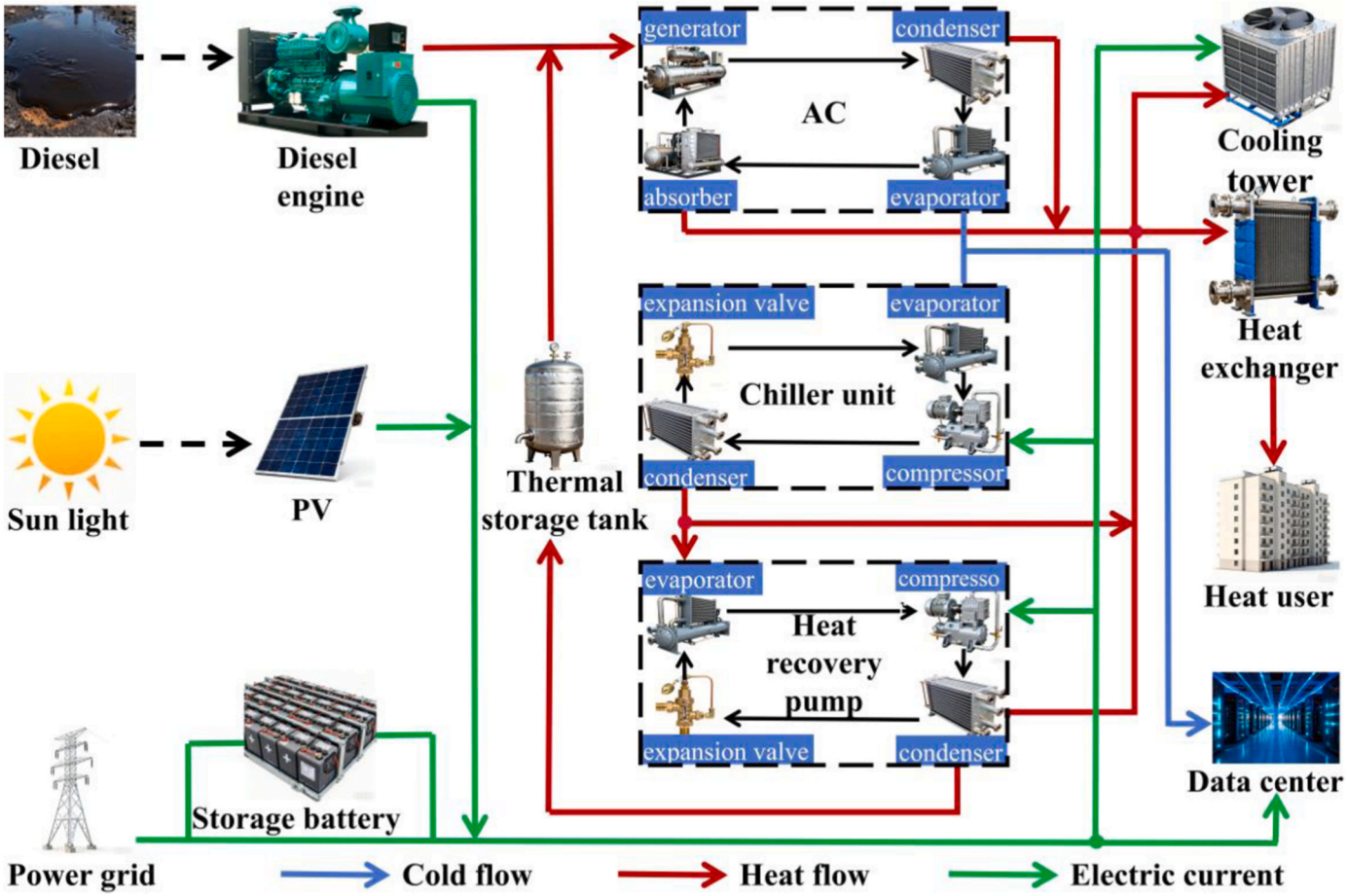


Fig. 1. Integrated energy system diagram of the data center.

incorporated to provide short-term temporal flexibility. On the demand side, the computational workload of the data center is classified into delay-sensitive and delay-tolerant tasks, enabling workload shifting within acceptable service quality constraints. Meanwhile, the thermal inertia of the data hall allows limited indoor temperature variations, further contributing to system flexibility.

In the proposed integrated energy system, the waste heat generated by IT equipment is not converted back into electricity due to its relatively low-grade thermal characteristics. Although low-grade IT waste heat can theoretically be upgraded via heat pumps and then converted to electricity through organic Rankine cycle units, this scheme suffers severe energy loss: the electricity consumed by heat pumps for heat grade improvement far exceeds the power output generated by thermoelectric conversion, leading to poor overall energy efficiency. Instead, the low-temperature heat from IT equipment, together with the waste heat discharged from diesel generators and electric chillers, is collected and boosted by heat recovery pumps to upgrade the thermal energy level. The heat users defined in this paper refer to residential heating loads distributed around the data center, which matches the actual heating demand characteristics of cold regions in northern Hebei, and the complete waste heat recovery and utilization route of the proposed system is clarified as follows. In non-heating seasons, all upgraded waste heat is delivered to absorption chillers to drive cooling production. During heating seasons, part of the recovered high-grade waste heat is transported to the aforementioned residential heat users through heat exchangers to satisfy district heating demand, while the residual heat continues to supply absorption cooling equipment. This two-branch cascaded utilization structure avoids direct waste heat dissipation through cooling towers and significantly improves the overall multi-energy utilization efficiency as well as the cascade utilization efficiency of waste heat. Moreover, the northern Hebei data center studied in this paper has continuous cooling demand all year round and long-term residential heating demand in heating seasons; direct cascade utilization of upgraded waste heat for cooling and district heating eliminates the extra power consumption required by heat-to-power conversion equipment and delivers prominent economic and low-carbon benefits, which further explains why thermoelectric conversion links are not arranged in the proposed system.

Through the coordinated interaction of these components, the proposed DC–IES transforms the data center from a passive energy consumer into an active flexible resource within the integrated energy system. This system architecture provides the foundation for the optimization model developed in the subsequent sections.

2.2. *System equipment modeling*

As illustrated in Fig. 1, the proposed data center integrated energy system consists of diesel generators, chilled units, absorption chillers, heat recovery pumps, battery storage systems, thermal storage tanks, photovoltaic units, cooling towers, and heat exchangers. Based on the system configuration shown in Fig. 1, the mathematical models of the major components are established as follows.

The energy conversion characteristics of different devices are described using conversion coefficients, including coefficients of performance (COPs) and energy conversion efficiencies. These parameters are assumed to remain constant during the optimization horizon and are summarized in Table 2.

Diesel Generator Model. As an essential backup power device in data centers, the diesel generator is characterized by its high starting torque and strong fuel adaptability, providing not only reliable power

Table 1
Classification of batch loads.

Batch Load						
Category	Short task (S)	Medium task (M)	Long task (L)	Level 1 delay	Level 2 delay	Level 3 delay
Time Length	15 min	30 min	1 h	1 h	3 h	5 h
Percentage	60%	30%	10%	30%	60%	10%

output but also enabling efficient waste heat recovery. Although it remains idle during normal operation, maintenance costs continue to accrue, making it a typical underutilized resource. In this study, the diesel generator is transformed from a passive standby unit into an actively dispatchable distributed power source, thereby fully exploiting its potential value. Assuming constant mechanical and combustion efficiencies, the generator's energy input, electrical output, and waste heat can be calculated using Equations (1)–(3):

$$Q_{in,d}(t) = m_d(t) LHV_d \tag{1}$$

$$P_d(t) = \eta_{e,d} Q_{in,d}(t) \tag{2}$$

$$Q_{ws,d}(t) = \eta_{ws,d} Q_{in,d}(t) \tag{3}$$

where $Q_{in,d}(t)$ represents the instantaneous energy input to the diesel generators (kW); $m_d(t)$ represents the mass flow rate of diesel (kg/s), indicating the amount of diesel consumed per second; LHV_d represents the lower heating value of diesel (kJ/kg), indicating the energy released when 1 kg of diesel is completely combusted; Δt represents the time interval (s), used to calculate the time period for energy input; $P_d(t)$ represents the electricity output from the diesel generators (kW); $\eta_{e,d}$ represents the power generation efficiency of the diesel generator; $\eta_{ws,d}$ represents the waste heat recovery efficiency; and $Q_{ws,d}(t)$ represents the output of waste heat (kW).

Chiller Unit Model. The chiller unit is a common cooling device in data centers, driven by electrical energy to produce cooling output, accompanied by the generation of waste heat. Assuming a constant coefficient of performance (COP), the output cooling capacity and waste heat can be calculated using the following formulas:

$$Q_c(t) = COP_c P_c(t) \tag{4}$$

$$Q_{ws,c}(t) = P_c + Q_c(t) \tag{5}$$

where $P_c(t)$ represents the instantaneous electrical energy input to the chiller unit (kW);$Q_c(t)$ represents the output cooling capacity (kW); and $Q_{ws,c}(t)$ represents the output waste heat (kW).

Absorption Chiller Model. The absorption chiller operates as a heat-driven cooling device that converts low-grade waste heat into cooling capacity, enabling cascaded energy utilization within the data center integrated energy system. By coupling the heating and cooling subsystems, it reduces dependence on electricity-driven cooling. Flexible operation of the absorption chiller allows waste heat to partially substitute electric cooling demand, thereby strengthening the coupling among electricity, heating, and cooling subsystems and providing additional operational flexibility for coordinated DC–IES optimization. The absorption chiller is modeled as follows:

$$Q_{ac}(t) = COP_{ac} Q_{in,ac}(t) \tag{6}$$

$$Q_{ws,ac}(t) = \eta_{ws,ac} Q_{in,ac}(t) \tag{7}$$

where $Q_{ar}(t)$ represents the cooling capacity of the AC (kW); $Q_{in,ac}(t)$ represents the input heat to the AC (kW); $Q_{ws,ac}(t)$ represents the waste heat discharged by the AC (kW); and $\eta_{ws,ac}$ represents the waste heat coefficient of the AC.

Heat Recovery Pump Model. The heat recovery pump is a key device for waste heat recovery and the cascading utilization of energy. Its mathematical model is primarily described through thermodynamic properties and energy balance. The following is a detailed analysis:

$$Q_{hrp,out}(t) = COP_{hrp} P_{hrp}(t) \tag{8}$$

$$Q_{hrp,in}(t) = \left(COP_{hrp} - 1\right) \cdot P_{hrp}(t) \tag{9}$$

where $P_{hrp}(t)$ represents the electrical energy input to the heat recovery pump (kW); $Q_{hrp,in}(t)$ represents the input heat to the heat recovery pump (kW); and $Q_{hrp,out}(t)$represents the total output heat of the heat recovery pump (kW).

Cooling Tower Model. The cooling tower is used to discharge waste heat from the system, and its power consumption is proportional to the heat dissipation. Assuming the heat dissipation efficiency of the cooling tower is determined by the coefficient β, the relationship between its power consumption and total heat dissipation is as follows:

$$P_{ct}(t) = \beta \cdot Q_{ws,total}(t) \tag{10}$$

where $P_{ct}(t)$ represents the instantaneous power consumption of the cooling tower (kW); $Q_{ws,total}(t)$ represents the total waste heat of the system (kW);

Storage Battery Model. The state of charge (SOC) of the battery is used to quantify the percentage of remaining charge, and its calculation varies depending on the charging and discharging mode. Assuming the nominal capacity C of the battery is constant, the variation of SOC over time can be calculated using the following formula:

$$SOC(t + \Delta t) = SOC(t) + \frac{\eta_{ch} P_{ch} \Delta t}{C} \times 100\% \tag{11}$$

Table 2
Technical and economic parameters of equipment.

Equipment	Parameters	Value	Equipment	Parameters	Value
Diesel generator	C_d	2100 kW	Heat storage tank	η_{in}	95%
	$\eta_{e,d}$	0.45		η_{out}	98%
	$\eta_{ws,d}$	0.5	Battery storage	Initial SOC	20%
Absorption chiller unit	COP_{ar}	0.8		SOC Range	20%-90%
	C_{ac}	1000 kW		P_{ch}	888 kW
	$\eta_{ws,ac}$	1.8		P_{dis}	420 kW
	$Cost_{ac}$	CNY 1.2 million per unit		η_{ch}	90%
Chilled unit	C_c	8800 kW		η_{dis}	90%
	COP_c	6	PV	S_{std}	1 kW/m^2
Heat recovery pump	C_{hrp}	1000 kW		r	0.00485/°C
	COP_{hrp}	5		T_s	25°C
	$Cost_{hrp}$	CNY 1.5 million per unit		N_{PV}	1000 units
Cooling Tower	β	0.07		C_{PV}	600W
Heat exchanger	η_{hx}	0.8		$Cost_{PV}$	2000CNY/kW

$$SOC(t+\Delta t) = SOC(t) - \frac{P_{dis}\Delta t}{\eta_{dis} C} \times 100\% \quad (12)$$

where $SOC(t)$ represents the state of charge at the current moment; P_{ch} is the charging power (kW), indicating the rate of energy input; η_{ch} is the charging efficiency, reflecting energy conversion losses; C is the nominal capacity of the battery (kWh), representing the maximum stored energy; P_{dis} is the discharging power (kW), indicating the rate of energy output; and η_{dis} is the discharging efficiency, reflecting energy output losses.

Thermal Storage Tank Model. The thermal energy storage tank (HTST) is a key thermal energy storage device. The state of the stored heat in the tank at any given moment is determined by the stored heat from the previous moment, the current charging heat, and the discharging heat. The dynamic process of this change can be described by the following energy balance equation:

$$HTST(t) = HTST(t-1) + \eta_{in} \cdot Q_{h,in}(t)\Delta t - \frac{Q_{h,out}(t)\Delta t}{\eta_{out}} \quad (13)$$

where $HTST(t)$ represents the amount of heat stored in the thermal energy storage tank at time (t)(kJ); $HTST(t\text{-}1)$ represents the amount of heat stored in the thermal energy storage tank at the previous time moment (t-1) (kJ); $Q_{h,in}(t)$ represents the heat input to the thermal energy storage tank at time (t) (kW); $Q_{h,out}(t)$ represents the heat released by the thermal energy storage tank at time (t) (kW); η_{in} is the charging efficiency; and η_{out} is the discharging efficiency.

Heat Exchanger Mode. During the heating season, the system efficiently recovers the waste heat generated by the absorption chiller (AC) and heat recovery pump through the heat exchanger, converting it into thermal energy for use by surrounding residents. This operational strategy not only significantly enhances the overall energy efficiency of the system and reduces energy waste but also enables cascading energy utilization during peak heating demand periods, thereby improving the system's economic and environmental benefits.

$$Q_{hx,out}(t) = Q_{hx,in}(t)\eta_{hx} \quad (14)$$

where $Q_{hx,out}(t)$ represents the output heat from the heat exchanger (kW); $Q_{hx,in}(t)$ represents the input heat to the heat exchanger (kW);

Photovoltaic Array Model. The photovoltaic array is the core power generation unit in the system that directly converts solar energy into electrical energy. Its output power exhibits significant intermittency and fluctuation. The output power of the photovoltaic panels primarily depends on the installed capacity, solar irradiance at the location, and environmental temperature. The power generation can be calculated using the following empirical model:

$$E_{PV} = P_{PV} N_{PV} \cdot \Delta t \quad (15)$$

$$P_{PV} = C_{PV} \times \frac{S}{S_{std}} \times \eta_{loss} \times \eta_{PV} \quad (16)$$

$$\eta_{loss} = 1 - r(T_{cell} - 1) \quad (17)$$

$$T_{cell} = 1.25(T_s - 20)S \quad (18)$$

where E_{PV} represents the power generation of the photovoltaic system (kJ); N_{PV} represents the number of photovoltaic panels installed (units), which is determined based on the maximum available installation area on the data center's roof or site to maximize the utilization of clean energy; P_{PV} represents the actual output power of a single photovoltaic panel (kW); C_{PV} represents the rated power of the photovoltaic panel (kW), i.e., the maximum output capacity under standard conditions; S represents the real-time irradiance (kW/m^2), reflecting the current light intensity; S_{std} represents the standard irradiance (kW/m^2); η_{loss} is the temperature loss coefficient; η_{PV} is the inverter efficiency; r is the standard temperature coefficient, indicating the proportion of power loss per 1°C increase in temperature; T_{cell} is the photovoltaic cell temperature (°C); T_s is the standard environment temperature (°C).

2.3. Data center flexible resource model

Data centers inherently possess multi-dimensional flexibility that can be actively exploited to support the coordinated operation of integrated energy systems. In this study, data center flexibility is represented from two complementary perspectives: computational workload flexibility and air thermal inertia. These two types of flexibility operate on different time scales and energy domains, jointly enabling dynamic coupling between electricity consumption, cooling demand, and system operation costs [16].

Specifically, workload flexibility allows part of the IT load to be temporally shifted while maintaining quality-of-service (QoS) requirements, transforming traditionally rigid electricity demand into a controllable flexible resource [17]. Meanwhile, the thermal inertia of indoor air provides short-term buffering capability for cooling demand, enabling time-shifted cooling operation without violating temperature constraints [18,19]. By jointly modeling these two mechanisms, the data center is transformed from a passive energy consumer into an active participant in multi-energy system optimization [20].

As illustrated in Fig. 2, computational load shifting primarily affects the electrical domain through peak shaving and valley filling, whereas air thermal inertia introduces flexibility in the thermal domain by decoupling instantaneous cooling demand from heat generation. The coordination of these flexible resources across multiple time scales significantly enhances the economic efficiency and operational resilience of the integrated energy system.

Data Center Load Shifting. Data center workloads can be categorized into interactive loads and batch processing loads according to their latency sensitivity and service characteristics. Interactive loads, such as online transactions and real-time computing tasks, require immediate processing and strict delay constraints. Any scheduling delay or migration may directly degrade QoS and user experience. Therefore, interactive loads are treated as inflexible demands and must be executed at their arrival time slots. In contrast, batch processing loads—including data backup, scientific computing, and model training—exhibit higher tolerance to execution delays. These tasks can be postponed or rescheduled within allowable time windows without affecting service quality, making them the primary source of computational flexibility in data centers.

assumptions. To simplify the research model and maintain generality, the following assumptions are made:

All servers in the data center are identical in terms of computational performance and power consumption characteristics, and the CPU real-time utilization rate can be used as an effective indicator of the system's operating state.

All computational tasks can be processed in parallel by multiple servers, and the system evenly distributes the total computational demand across all currently active servers [21].

In each scheduling period, the initial number of tasks arriving at the data center follows an independent and identically distributed pattern.

During the scheduling cycle, the processing rate of each server remains constant.

Definition of the Data Center Computational Task Scheduling Model. Time Slot Set: The set of time slots within the scheduling cycle is defined as $T = \{t_1, t_2, \ldots, t_{24}\}$, where each time slot has a length of $\Delta t = 1$ h, covering a total of 24 h.The scheduling horizon is set to one day and discretized into hourly intervals. Therefore, (T = 24) in all simulation scenarios.The set of servers in the data center is represented as $N = \{n_1, n_2, \ldots, n_m\}$, where m is the total number of servers.In time slot t_i, the computational tasks arriving at the data center are random and are categorized into three types based on their average processing duration: small tasks (S), with a processing time of $lS = 0.25\Delta t$ (15 min); medium tasks (M), with a processing time of $lM = 0.5\Delta t$ (30 min); and large tasks (L), with a processing time of $lL = \Delta t$ (1 h).Based on the allowable

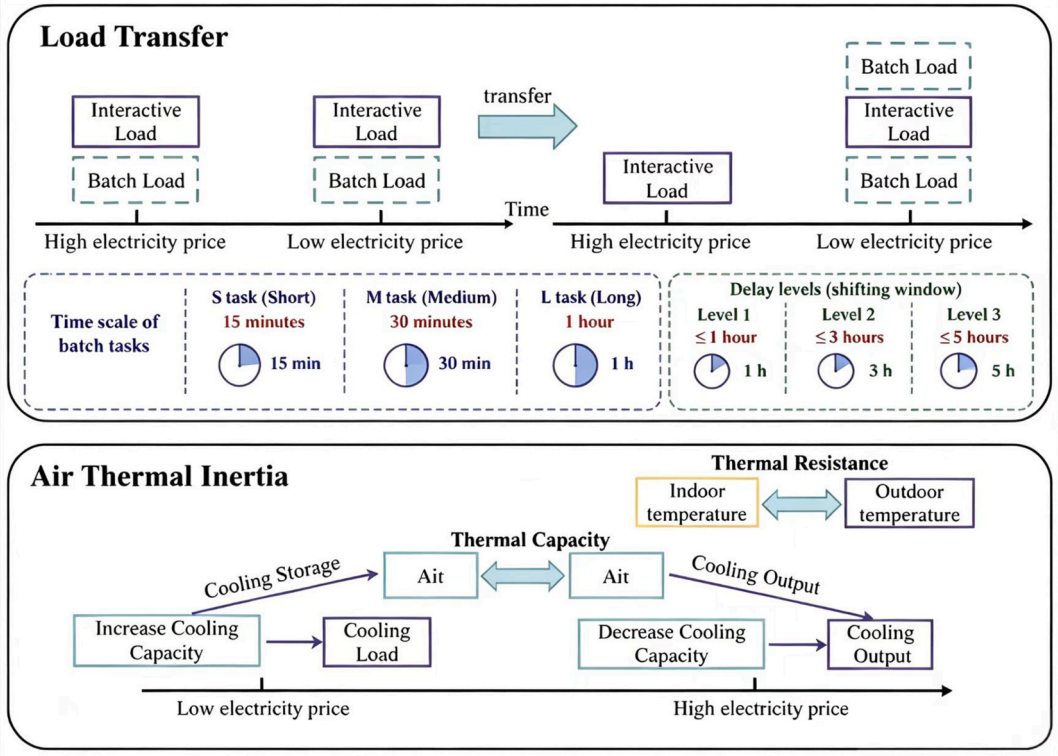


Fig. 2. Schematic diagram of data center flexible resource characteristics.

execution delay of the tasks, they are further classified into four levels: Level 0 tasks must be executed immediately and cannot be delayed; Level 1 tasks can be delayed until the next time slot (maximum delay $1\Delta t$); Level 2 tasks can be delayed up to three time slots (maximum delay $3\Delta t$); and Level 3 tasks can be delayed up to five time slots (maximum delay $5\Delta t$).

The data shown in Table 1 is based on the scheduling characteristics of actual workloads. Its aim is to improve the resource utilization efficiency of the data center through reasonable load distribution strategies and to provide practical references for load shifting and optimized scheduling.

Load Shifting Matrix Model. Since computational tasks can only be shifted from the arrival time slot to subsequent time slots, and cannot be shifted to previous time slots, the load shifting coefficient matrix is an upper triangular matrix.

The matrix naming rule is $X^{PT,DL}$, where X represents the shifting matrix, PT denotes the processing duration, and DL indicates the delay level. For example, $X^{S,1}$ represents the shifting matrix for tasks with a processing duration of S and a delay level of 1, while $X^{L,3}$ represents the shifting matrix for tasks with a processing duration of L and a delay level of 3. The shifting coefficient matrix model is as follows:

$$X^{PT,DL} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,24} \\ 0 & x_{2,2} & \cdots & x_{2,24} \\ 0 & \ddots & \cdots & \vdots \\ 0 & 0 & 0 & x_{24,24} \end{pmatrix} \tag{19}$$

$$PT \in \{S, M, L\} \quad DL \in \{0, 1, 2, 3\} \tag{20}$$

The shifting coefficient $X_{m,n}$ represents the percentage of task volume shifted from time slot m to time slot n relative to the total task volume in time slot m. It is used to quantify the relative scale of load scheduling between different time slots.

The initial task volume arriving at the data center during each time slot within an optimization period can be expressed in matrix form as follows:

$$N_0^{PT} = \left(n_{0,1}, n_{0,2} \cdots n_{0,t}, n_{0,24}\right) \tag{21}$$

where *NPT 0*represents the total number of computational tasks initially arriving within a day, and $n_{0,t}$ denotes the number of computational tasks arriving at time slot t.

After applying the shifting coefficients to shift the task load, the number of computational tasks starting execution in time slot n within the data center is expressed as:

$$\begin{aligned} N_n^{PT} &= N_0^{PT} \cdot X^{PT,DL} \\ &= \left(n_{0,1}^{PT} \quad n_{0,2}^{PT} \cdots n_{0,24}^{PT}\right) \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,24} \\ 0 & x_{2,2} & \cdots & x_{2,24} \\ 0 & \ddots & \cdots & \vdots \\ 0 & 0 & 0 & x_{24,24} \end{pmatrix} \end{aligned} \tag{22}$$

where *NPT n* represents the number of computational tasks starting execution in time slot n after applying the shifting coefficients to shift the task load.

Based on the above task allocation strategy and server operation assumptions, the energy consumption of the data center's IT equipment can be described using a linear model. Specifically, the energy consumption at any given time ttt can be expressed as:

$$P_{IT}(t) = \left(P_{IT,max} - P_{IT,base}\right) \cdot u_{CPU}^t + P_{IT,base} \tag{23}$$

where $P_{IT}(t)$ represents the total energy consumption of the IT equipment in the data center at time t (kW), $P_{IT,max}$ is the maximum energy consumption when all servers are at full load, $P_{IT,base}$ is the baseline energy consumption when all servers are idle, and *ut CPU* represents the average CPU utilization of the servers at time t.

The CPU utilization of the server reflects the usage of computational resources and is calculated based on the ratio of the task computational load to the total available resources of the server.

$$u_{CPU}^t = \frac{\sum_{PT \in \{S,M,L\}} N_t^{PT} \cdot PT}{m \cdot \Delta t} \tag{24}$$

where N_t^{PT} represents the number of tasks of type *PT* in time slot *t*, *PT* is the computational load of a single task, and *m* denotes the total number of servers (units).

Data Center Thermal Inertia. Data center room air thermal inertia is primarily driven by the thermal capacity of the large volume of air within the room. This characteristic ensures that when there is a temporary increase in thermal load (such as brief high load on some equipment) or when the cooling system needs to be adjusted, the internal temperature does not immediately exceed the safety threshold. Instead, it can temporarily maintain or slowly rise within an acceptable range, thus providing a buffer time for the cooling system to adjust. This helps avoid frequent system starts due to minor thermal fluctuations and allows for a reduction in unnecessary cooling operations by reasonably utilizing the temperature buffer period. Ultimately, this enables flexible regulation of the data center's thermal load.The thermal inertia model for data centers can be expressed using an equivalent thermal parameter model, which can be represented as a first-order linear homogeneous differential equation [22,23]. After simplification, it can be written as:

$$\begin{cases} \left(T_t^{out} - T_t^{in}\right) H^{DC} S^{DC} + Q_{ws}^t - Q_{cs}^t = \rho_a C_a V^{DC} \delta_t^{in} \\ T_t^{in} - T_0^{in} = \delta_t^{in} \Delta t, t = 1 \\ T_t^{in} - T_{t-1}^{in} = \delta_t^{in} \Delta t, 2 \le t \le 24 \\ Q_{cs}^t \ge 0 \\ T_{\min}^{in} \le T_t^{in} \le T_{\max}^{in} \end{cases} \tag{25}$$

where *Tout t*,*Tin t* and $\delta in\ t$ represent the outdoor temperature (°C), indoor temperature (°C), and the rate of change of indoor temperature, respectively; H^{DC},S^{DC}andV^{DC}represent the heat dissipation coefficient (W/m^2·K), heat exchange area (m^2), and volume (m^3) of the data center, respectively; *Qt ws* and *Qt cs* represent the waste heat power (kW) and the required cooling power (kW) generated by the data center, respectively; and ρ_a and C_a represent the air density (kg/m^3) and specific heat capacity (J/kg·K), respectively.The term $H_{DC}S_{DC}$(*Tt out-Tt in*) represents the heat transfer between the server room and the outdoor environment through the building envelope, including walls and roofs. Ventilation heat transfer is neglected in this study because modern data centers generally operate under highly enclosed conditions with limited outdoor air exchange.*Qt ws* denotes the waste heat generated by IT equipment and other electrical devices. Since almost all electrical energy consumed by servers is eventually converted into heat, *Qt ws* is treated as an internal heat source and is assumed to be proportional to the IT load. *Qt cs* represents the cooling capacity supplied by the cooling system.

3. System optimization model and solution method

Based on the data center IES constructed in Chapter 2 and the in-depth analysis of IT load shifting characteristics and building thermal inertia characteristics, this chapter aims to establish a mathematical programming model with the objective of minimizing the system's daily operating total cost and ensuring service quality. The model simultaneously takes into account the system's economic operation and the service quality of computational tasks. Its constraint system comprehensively includes power balance constraints for electricity, cooling, and heating, operational constraints for energy conversion equipment, and spatiotemporal load shifting constraints for IT loads. Ultimately, it is formulated as a Mixed-Integer Linear Programming (MILP) problem [24–26].

3.1. Objective function

Minimize the operating cost. The primary optimization objective is to minimize the total system operating cost over the scheduling period. Its mathematical expression is as follows:

$$F_1 = \min\left(Fuel_{\mathrm{Cos}\ t} + Grid_{\mathrm{Cos}\ t} - Heat_{Value}\right) \tag{26}$$

where $Fuel_{Cost}$ represents the diesel cost, $Grid_{Cost}$ represents the electricity purchase cost, and $Heat_{Value}$ represents the revenue generated from heating supplied to residents during the heating season (with heat coming from the AC and heat recovery pump). The specific calculation formulas for these three are as follows:

$$Fuel_{\mathrm{Cos}\ t} = \left(\frac{\rho_{fuel}}{\frac{LHV_d}{3.6}\cdot\eta_{e,d}}\right)\sum_{t=1}^{T} E_d(t) \tag{27}$$

$$Grid_{\mathrm{Cos}\ t} = c_{grid}(t)\sum_{t=1}^{T} E_{grid}(t) \tag{28}$$

$$Heat_{Value} = V_{ng}\cdot\rho_{ng} \tag{29}$$

$$V_{ng} = \frac{3.6\cdot\left(Q_{ws,ac} + Q_{hrp,out}\right)}{Q_{ng}\cdot\eta} \tag{30}$$

where ρ_{fuel} represents the diesel price (CNY/kg), $c_{gird}(t)$ represents the real-time electricity price, $E_{gird}(t)$ represents the total electricity purchase from the grid at time *t*, V_{ng} refers to the volume of gas required (m^3) to fully replace the data center's waste heat with a gas boiler to meet the same heating demand, Q_{ng} represents the heating value of the gas combustion, taken as 36.44 MJ/m^3, ηrepresents the gas boiler efficiency, used to quantify the effective utilization ratio of the boiler for gas heat, set at 0.93, and ρ_{ng} represents the natural gas price, taken as 3.2 CNY/m^3.

Maximize the service quality. In the load shifting optimization model, in addition to minimizing the system operating cost, the user service quality requirements must also be taken into account. This model uses the average service quality of batch processing tasks across each time slot in the scheduling period as an indicator to measure the overall service level of the data center. A higher service quality indicates shorter task delays and lower frequency of delays; conversely, lower service quality reflects longer overall task delays and more frequent delays. The service quality is calculated as:

$$F_2 = \max\ \sum_{t=1}^{24} Q_{batch,t} \tag{31}$$

$$Q_{batch,t} = \frac{1}{N_{batch,t}}\sum_{k\in batch}\left(1 - \frac{DL_k}{DL_{k,max}}\right) \tag{32}$$

where $Q_{batch,t}$ represents the average service quality of all batch processing tasks during time slot *t*, $N_{batch,t}$ denotes the total number of batch processing tasks in time slot *t*, DL_k represents the actual delay time of the k-th batch processing task, and $DL_{k,max}$ denotes the maximum allowable delay time for the k-th batch processing task.

3.2. Constraints

Energy Balance Constraints.

Power Balance Constraints. The electrical load of the system during the non-heating season consists of the power consumption of IT equipment, chiller units, heat recovery pumps, and cooling towers. The power supply comes from the diesel generators, photovoltaic system, external grid, and battery discharge power, which collectively meet the demand. The instantaneous energy relationship of the system is expressed by the following equation:

$$P_{grid}(t) + P_d(t) + P_{PV}(t) + P_{dis}(t) = P_c(t) + P_{hrp}(t) + P_{ct}(t) + P_{IT}(t) + P_{ch}(t) \tag{33}$$

where $P_{PV}(t)$ represents the instantaneous power generation of the photovoltaic system(kW), $P_{dis}(t)$ is the instantaneous discharge power of the battery,(kW), and $P_{ch}(t)$ is the instantaneous charging power of the battery, (kW).

During the heating season, the system operates without the cooling tower, resulting in a corresponding reduction in the electrical load. The instantaneous power balance relationship of the system is expressed by the following equation:

$$P_{grid}(t) + P_d(t) + P_{PV}(t) + P_{dis}(t) = P_c(t) + P_{hrp}(t) + P_{IT}(t) + P_{ch}(t) \quad (34)$$

Cooling Load Balance Constraints. During both the non-heating and heating seasons, the cooling load of the data center is generated by the heat dissipation from IT equipment. The cooling supply is jointly provided by the AC and chiller units. The instantaneous cooling load balance relationship of the system is expressed by the following equation:

$$Q_c(t) + Q_{ac}(t) = Q_{IT}(t) \quad (35)$$

where $Q_{IT}(t)$ represents the cooling load of the data center equipment at time t(kW).

Heat Load Balance Constraints. The heat balance of the system during the non-heating season is based on the internal waste heat utilization. When the electricity price is low, the waste heat from the chiller units is recovered by the heat recovery pump and stored in the thermal storage tank. When the electricity price is high, the heat from the thermal storage tank and the waste heat from the diesel generator jointly supply the AC. The waste heat from the AC and the chiller units is eventually processed by the cooling tower. The instantaneous heat balance relationship of the system is expressed by the following equation:

$$Q_{ws,c,hrp}(t) = Q_{hrp,in}(t) \quad (36)$$

$$Q_{hrp,out}(t) = Q_{h,in}(t) \quad (37)$$

$$Q_{in,ac}(t) = Q_{ws,d}(t) + Q_{h,out}(t) \quad (38)$$

$$Q_{ws,total}(t) = Q_{ws,c,ct}(t) + Q_{ws,ac}(t) \quad (39)$$

where $Q_{ws,c,hrp}(t)$represents the portion of the waste heat from the chiller units that enters the heat recovery pump (kW), and $Q_{ws,c,ct}(t)$ represents the portion of the waste heat from the chiller units that enters the cooling tower (kW).

During the heating season, the waste heat from the diesel generator is directly supplied to the AC, and the heat recovery pump recovers all the waste heat from the chiller units. The heat output from the heat recovery pump, together with the waste heat from the AC, is supplied to the heat exchanger for heating the users. The heat balance relationship of the system is expressed by the following equation:

$$Q_{in,ac}(t) = Q_{ws,d}(t) \quad (40)$$

$$Q_{ws,c}(t) = Q_{hrp,in}(t) \quad (41)$$

$$Q_{hx,in}(t) = Q_{hrp,out}(t) + Q_{ws,ac}(t) \quad (42)$$

Equipment Constraints. To ensure the safe and stable operation of the system, the power of each key equipment must remain within its rated operating range. The specific constraints are as follows:

$$P_{\min,i} \le P_i(t) \le P_{\max,i} \quad (43)$$

where $P_{\min,i}$ represents the lower output limit of the i -th equipment (kW); $P_{\max,i}$ represents the upper output limit of the i -th equipment (kW); P_i(t) represents the output of the i -th equipment at time t (kW); and i refers to equipment within the system, including the diesel generator, chiller unit, AC, heat recovery pump, cooling tower, etc.

Battery Constraints. The operation of this battery model must satisfy power and state-of-charge (SOC) constraints. Both charging and discharging power must be non-negative and cannot exceed the maximum power limits. The state of charge (SOC) must always remain between the allowable minimum and maximum values. The specific expressions are as follows:

$$\begin{cases} 0 \le P_{ch}(t) \le P_{ch,max} \\ 0 \le P_{dis}(t) \le P_{dis,max} \\ SOC_{min} \le SOC(t) \le SOC_{max} \end{cases} \quad (44)$$

Where $P_{ch,max}$ represents the maximum charging power (kW); $P_{dis,max}$ represents the maximum discharging power (kW); SOC$_{min}$ represents the minimum state of charge; and SOC$_{max}$ represents the maximum state of charge.

Thermal Storage Tank Constraints. The operation of the thermal storage tank must satisfy constraints on capacity, power, and mode. The stored heat must remain within the rated range; both charging and discharging power must be non-negative and within the limits; and the charging and discharging modes must be mutually exclusive, achieved through binary variables. The specific expressions are as follows:

$$\begin{cases} 0 \le HTST(t) \le HTST_{\max} \\ 0 \le Q_{h,in}(t) \le Q_{h,in}^{\max} \\ 0 \le Q_{h,out}(t) \le Q_{h,out}^{\max} \\ Q_{h,in}(t) \le u_{in}(t) \cdot Q_{h,in}^{\max} \\ Q_{h,out}(t) \le u_{out}(t) \cdot Q_{h,out}^{\max} \\ u_{in}(t) + u_{out}(t) \le 1 \quad u_{in}(t), u_{out}(t) \in \{0,1\} \end{cases} \quad (45)$$

where $HTST_{max}$ represents the maximum heat storage capacity of the thermal storage tank (kJ); $Qmax\ h,in$ represents the maximum charging power (kW); $Qmax\ h,out$represents the maximum discharging power (kW); u is a set of binary decision variables used to control the operation modes of the thermal storage tank. u_{in}(t) = 1 indicates the activation of the charging mode, and u_{out}(t) = 1 indicates the activation of the discharging mode, with mutual exclusivity ensuring the safe operation of the system.

Load Shifting Coefficient Constraints. Shifting Coefficient Range: The shifting coefficient must lie between 0 and 1. For all PT-DL tasks arriving at time slot m, they must be assigned to an allowed time window starting from m (the sum of the shifting coefficients must equal 1). If $n < m$, or if n exceeds the allowed time window, the shifting coefficient is equal to zero.

$$0 \le x_{m,n}^{PT,DL} \le 1 \quad (46)$$

$$\begin{cases} \sum_{n=m}^{\min\left(m+DL_{\max},24\right)} x_{m,n}^{PT,DL} = 1 \\ x_{m,n}^{PT,DL} = 0 \quad if\ n > m + DL_{\max}\ or\ n < m \end{cases} \quad (47)$$

where DL_{max} represents the maximum allowable delay time for the batch processing tasks.

Load Shifting Matrix Constraint. The delay level for interactive tasks is 0. These tasks typically require real-time processing and cannot be shifted, so they must remain on the original node m for immediate execution and cannot be forwarded to other nodes n. Batch tasks (with delay levels 1, 2, or 3) must either be shifted or retained. These tasks allow for some delay, so they need to be executed on the current node mmm or on a node within a limited subsequent range (such as nodes from $m+1$ to $m + DL_{max}$.

$$X_{m,n}^{PT,0} = 0 \ \ \forall\, n \Big/ m \quad (48)$$

$$\sum_{n=m}^{m+DL_{\max}} X_{m,n}^{PT,1,2,3} = 1 \quad (49)$$

where $XPT,0\ m,n$represents the interactive load task matrix, and $XPT,1,2,3\ m,n$represents the batch load task matrix.

CPU utilization constraint. To ensure system reliability and prevent overload, the total server utilization for each time period t must be strictly limited to within the maximum allowed value of 90%.

$$0 \le u_{CPU}^{t} \le 90\% \tag{50}$$

3.3. Evaluation metrics

In order to scientifically and comprehensively assess the overall benefits of the optimization strategy, this study evaluates it from three dimensions: economic efficiency, environmental benefits, and investment feasibility. Using the traditional sub-supply system as a reference benchmark, three core indicators—cost savings rate (CSR), carbon emission reduction rate (ERR), and payback period (PBP)—are selected to jointly construct a comprehensive evaluation system for the data center integrated energy system.

Economic evaluation metrics. To reflect the economic efficiency of the integrated energy system relative to the reference system, the annual savings rate is used to evaluate the system's economic performance. The specific expression is as follows:

$$CSR = \frac{\mathrm{Cos}\, t_{trad} - \mathrm{Cos}\, t_{opt}}{\mathrm{Cos}\, t_{trad}} \times 100\% \tag{51}$$

where $Cost_{trad}$ represents the total operating cost of the data center under the traditional model (CNY/year); $Cost_{opt}$ represents the total operating cost after optimization (CNY/year).

Environmental evaluation metrics. In order to compare the environmental benefits of the integrated energy system relative to the reference system, the carbon dioxide emission reduction rate is selected as the environmental metric for the system. The calculation method is as follows:

$$ERR = \frac{C_{trad} - C_{opt}}{C_{opt}} \times 100\% \tag{52}$$

$$C_{out} = \sum_{t=1}^{T} \left(\alpha_{grid} \cdot E_{grid} + \alpha_{ng} \cdot V_{ng}\right) \tag{53}$$

where C_{trad} represents the total carbon dioxide emissions under the traditional model (kg); C_{opt} represents the total carbon dioxide emissions of the optimized system (kg); α_{grid} is the carbon emission factor of grid electricity ($kgCO_2/kWh$), used to quantify the indirect carbon emissions from purchased electricity; α_{ng} is the carbon emission factor of natural gas ($kgCO_2/m^3$), used to calculate the direct carbon emissions from the combustion of natural gas in gas equipment.

Payback period. This study involves an integrated energy system composed of both existing traditional equipment and newly added key technological devices. The new devices include AC, heat recovery pumps, and photovoltaic systems, which are the incremental investment subjects of this technological renovation. The system coordinates the resources of both new and old equipment to maximize the utilization of waste heat and solar energy, thereby generating savings in operating costs. The payback period is then calculated based on these savings to evaluate its economic efficiency. The specific calculation formula is as follows:

$$PBP = \frac{\Delta CAPEX}{\Delta OPEX} \tag{54}$$

$$\Delta CAPEX = \mathrm{Cos}\, t_{ar} + \mathrm{Cos}\, t_{hrp} + \mathrm{Cos}\, t_{PV} \tag{55}$$

$$\Delta OPEX = \mathrm{Cos}\, t_{trad} - \mathrm{Cos}\, t_{opt} \tag{56}$$

where $\triangle CAPEX$ represents the total initial investment cost for implementing the optimized strategy described in this project $Cost_{ac}$, $Cost_{hrp}$ and $Cost_{PV}$ represent the initial investment costs of AC, heat recovery pumps, and photovoltaic panels, respectively (CNY); $\triangle OPEX$ refers to the reduction in annual operating costs of the system compared to the baseline scenario after adopting the optimized strategy (CNY/year).

4. Case study

4.1. Basic parameters

This case study is based on real operational data from a large-scale data center located in Hebei Province, China. Hourly cooling and electrical load data for the year 2024 are employed to represent realistic operating conditions. To capture representative daily operational characteristics while ensuring computational tractability, a typical daily load profile is constructed using clustering techniques calibrated against the annual total load level of the data center [27]. The resulting typical daily load demand, together with the corresponding variations in cooling and electrical loads, is illustrated in Fig. 3. In this study, batch tasks are assumed to arrive according to the typical daily workload profile shown in Fig. 3. The arrival volume in each time slot is predefined based on historical operating data. Therefore, the optimization focuses on determining the execution time of flexible tasks rather than modeling stochastic task arrivals.The technical and economic parameters of the equipment in the integrated energy system are selected based on existing literature and actual engineering data, and are summarized in Table 2. The time-of-use electricity pricing adopted in this study is obtained from the electricity tariff standards published by the South Hebei Power Grid. Seasonal price differences between the heating and non-heating periods are explicitly considered, as shown in Fig. 4. The key parameters of the data center are provided in Table 3.

4.2. Optimization result analysis

Scenario comparison analysis. To validate the effectiveness of the proposed data center model, we conducted simulation analyses of five different scenarios listed in Table 4. Among them, S1 represents the traditional scenario, with a simple and rigid system operation mode. In summer, this scenario relies entirely on electrically driven chilled units to meet the total cooling load of the data center, with waste heat generated during operation directly discharged to the atmosphere via cooling towers. In winter, a plate heat exchanger combined with a cooling tower is used to directly discharge the waste heat generated by the data center. In this scenario, the indoor temperature is strictly set and maintained at a constant 20°C, and all equipment operates in a fixed mode without any flexible adjustment capabilities. Scenarios S2, S3, and S4 each introduce one of the three key features of the built data center model: integrated energy systems, load shifting, and thermal inertia.

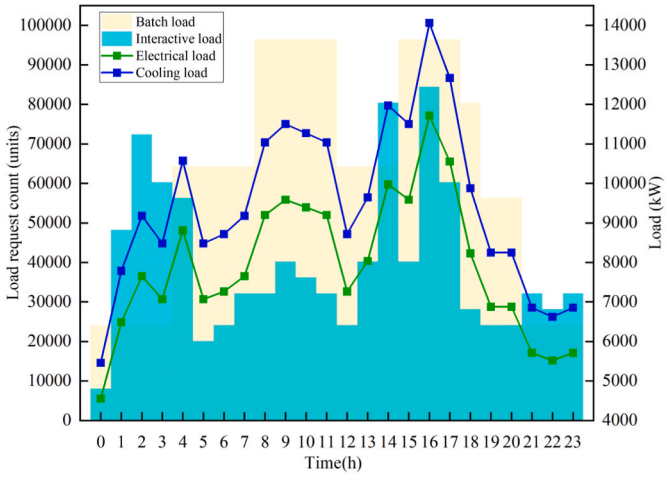


Fig. 3. Trends of typical daily load requests and cooling/electrical load variations in data centers.

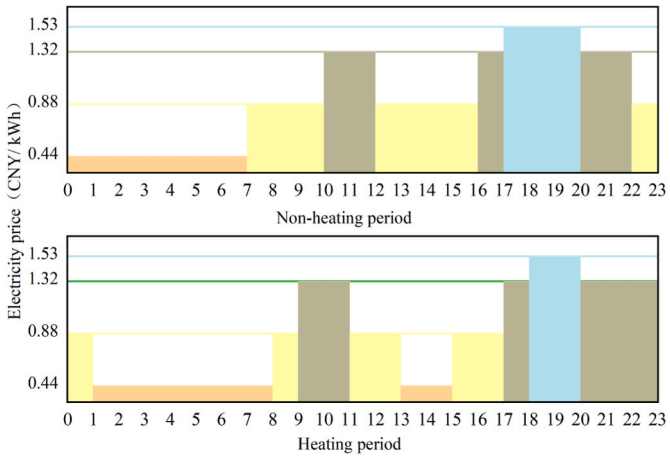


Fig. 4. Time-of-use electricity price chart for data center during heating and non-heating seasons.

Table 3
Data center parameters Table.

Parameters	Value	Parameters	Value
N	83000 units	V_{DC}	120000 m^3
$E_{IT,max}$	13000 kWh	S_{DC}	10000 m^2
$E_{IT,base}$	3000 kWh	*Max* ΔT *rate*	3 °C/h
ρ_{DC}	1.2 kg/m^3	*TDC,in min-TDC,in max*	16°C-25°C
C_{DC}	1005/(3.6e6) [kWh/kg/°C]	H_{DC}	1.25e-5 [kW/m^2/°C]

Table 4
Data center scenario settings.

Scenario	IES	Load shifting	Thermal inertia
S1	×	×	×
S2	✓	×	×
S3	×	✓	×
S4	×	×	✓
S5	✓	✓	✓

Scenario S5 represents the complete model proposed in this paper.

Table 5 presents the optimal equipment capacity configuration results, covering the capacities of key equipment such as the diesel generator, chilled unit, heat recovery pump, and AC. Table 6 shows the fuel cost, electricity purchase cost, heating revenue, and daily total cost for the data center under different operational scenarios, along with the cost savings compared to the baseline scenario S1 (unit: CNY). By combining the optimized scheduling results for a typical day, the rationality and effectiveness of the integrated energy system model for the data center proposed in this paper can be further validated. This demonstrates that considering flexible loads and thermal inertia characteristics in system planning is crucial for achieving energy savings, reducing consumption, and optimizing economics. Based on this, a detailed analysis of the operational results for each scenario will be further conducted.

Data center integrated energy system. As shown in Table 5, S1 represents the traditional energy supply mode, while S2 introduces an integrated energy system (IES), including AC and heat recovery pump devices. The output configuration of the diesel generator and chilled unit is more coordinated in S2, fully demonstrating the advantages of multi-energy collaboration. In the S2 scenario, the capacity of the chilled unit is not reduced because, during peak cooling load periods, the electricity price is lower, and the system tends to directly use the chilled unit for cooling, with the AC unit not participating in cooling during this

Table 5
Comparison of equipment capacity configuration for different data center scenarios.

Scenario	Period	Inherent equipment		Additional equipment	
		Diesel generator (kW)	Chilled unit(kW)	Heat recovery pump(kW)	Absorption chiller unit (kW)
S1	Non-heating	/	14063	/	/
	heating	/	0	/	/
S2	Non-heating	10630	11970	12339	14063
	heating	11625	14063	20505	10334
S3	Non-heating	/	14063	/	/
	heating	/	14063	/	/
S4	Non-heating	/	14063	/	/
	heating	/	14063	/	/
S5	Non-heating	10555	12342	12480	14063
	heating	11585	14063	20680	10298

period. The capacity design of the chilledunit is based on the maximum output demand of the day, so the capacity remains unchanged. Further analysis in conjunction with Table 6 shows that, during the non-heating season, the S2 scenario reduces daily operational costs by about 12,600 CNY compared to S1 through electricity price adjustments. In the heating season, heating revenue is achieved through waste heat recovery, and the total cost is reduced by about 59,600 CNY compared to S1, resulting in significant energy-saving and economic benefits.

Figs. 5–7 illustrate the energy balance and cooling supply structure under Scenario S2. During the non-heating season, the system actively exploits time-of-use electricity prices by charging batteries and operating electrically driven chillers during low-price periods, while relying on diesel generation and absorption cooling during high-price periods. Waste heat from both diesel generators and chilled units is effectively recovered via heat recovery pumps, reducing heat rejection losses and improving overall energy efficiency. During the heating season, recovered waste heat is prioritized to meet heating demand, generating additional heating revenue. These results demonstrate that integrated energy systems enable data centers to transform waste heat from a byproduct into an economic resource, enhancing both economic and energy performance.

Data center load shifting. Given the similarity of optimization outcomes between seasons, the non-heating season is selected for detailed analysis. A bi-objective optimization framework balancing operating cost and quality of service (QoS) is employed, and the Pareto frontier is shown in Fig. 8. In Fig. 8, the optimal scheme is selected by weighting method to construct the comprehensive optimization model. The weight coefficient of F_1 ($w = 0.75$) is optimized to achieve substantial cost reduction while maintaining acceptable service quality.

As shown in Tables 5 and 6, load shifting smooths the IT load profile through peak shaving and valley filling, leading to notable cost savings compared with Scenario S1. Although load shifting does not reduce total energy consumption, it improves the temporal alignment between energy demand and electricity prices, thereby enhancing economic performance without compromising QoS.

Fig. 9 further provides a visual comparison of costs and energy consumption before and after load shifting during the non-heating season, from both time and quantity perspectives. The figure clearly shows that the load shifting strategy successfully shifts some of the computing tasks that were originally scheduled during the peak electricity price periods to the lower electricity price off-peak periods, achieving coordinated optimization at both the system and equipment levels.

Fig. 10 further provides a detailed comparison of the execution of 9 types of batch tasks before and after optimization. The results show that

Table 6
Cost comparison for different data center scenarios.

Scenario	Period	Fuel cost	Electricity cost	Heating revenue	Daily total cost	Cost savings
S1	Non-heating	/	232348.23	/	232348.23	/
	heating	/	185091.67	/	185091.67	/
S2	Non-heating	42589.88	177152.15	/	219742.03	12606.2
	heating	98785.40	126250.13	103146.76	121888.77	63202.9
S3	Non-heating	/	223858.69	/	223858.69	8489.54
	heating	/	177378.57	/	177378.57	7713.1
S4	Non-heating	/	232191.66	/	232191.66	156.57
	heating	/	181449.32	/	181449.32	3642.35
S5	Non-heating	41313.23	174602.04	/	215915.27	16432.73
	heating	91050.95	129776.18	101998.19	118828.94	66262.73

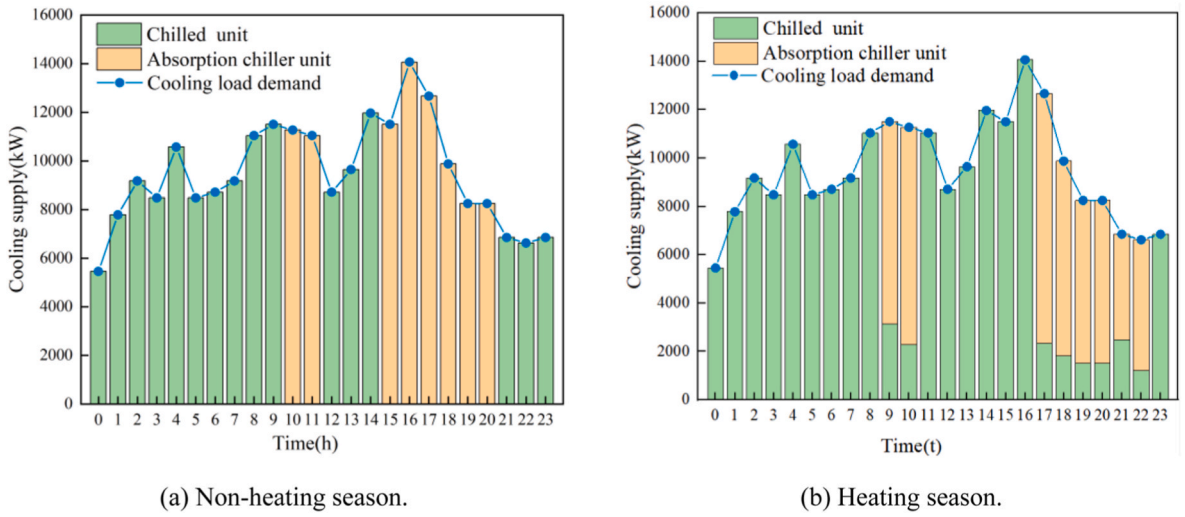


Fig. 5. Comparison of cooling supply structures in different seasons.

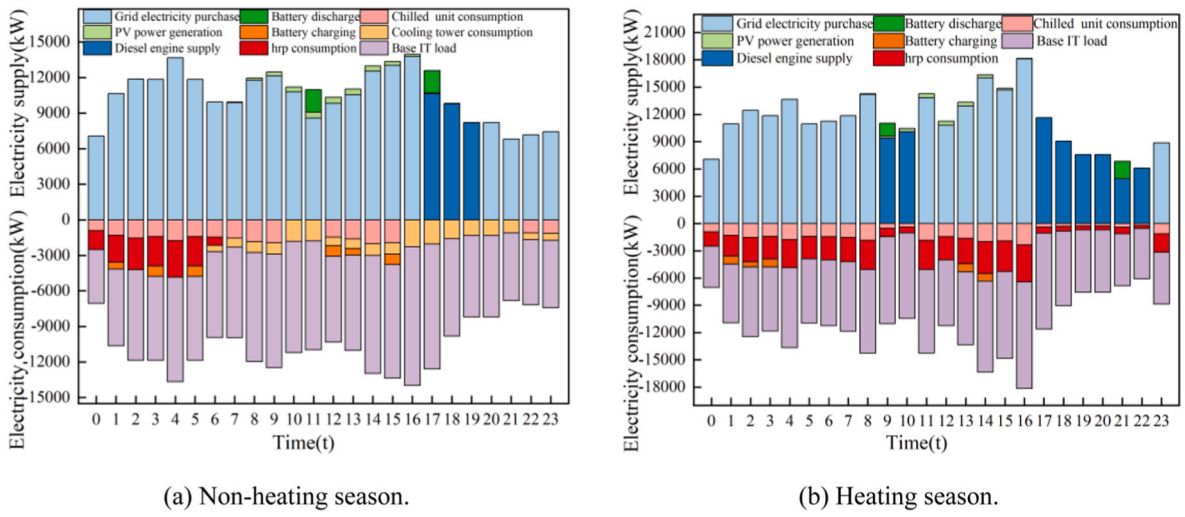
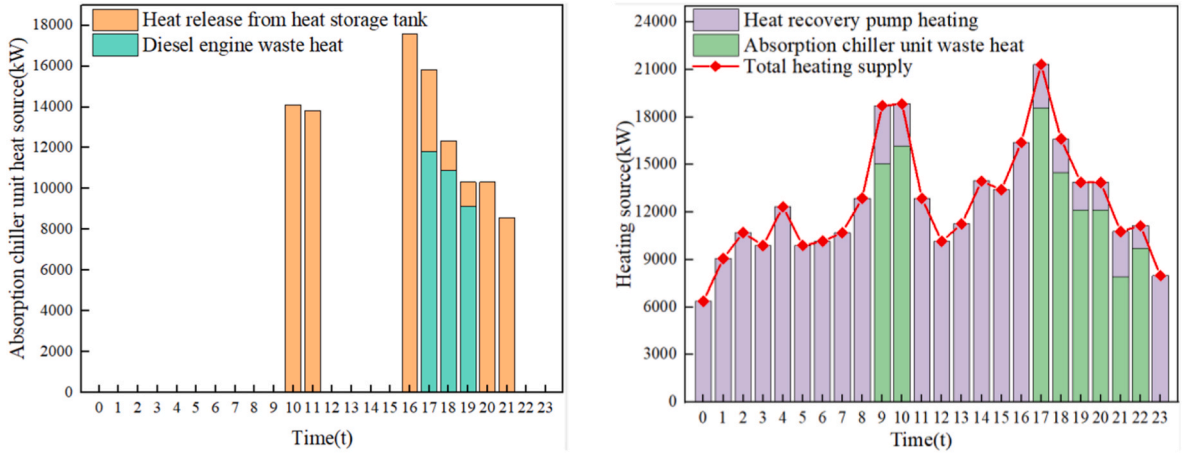


Fig. 6. Comparison of electricity balance in different seasons.

the load shifting strategy has varying optimization effects on different types of tasks: tasks with long cycles and high latency tolerance are efficiently scheduled to periods with lower energy costs, significantly reducing processing costs while meeting completion deadlines; on the other hand, delay-sensitive tasks are prioritized for immediate processing, ensuring the stability of overall service quality.

Data center air thermal inertia. The cost comparison results in Table 6 show that utilizing air thermal inertia can further reduce the daily operating costs of the data center. The fundamental source of this economic benefit lies in the fact that thermal inertia allows the indoor temperature of the server room to fluctuate dynamically within a safe range (16–25°C), providing flexible space for adjusting the cooling system's power, thus effectively reducing cooling energy consumption during high electricity price periods. As shown in Figs. 11 and 12, cooling power is significantly reduced during high electricity price periods through pre-cooling strategies implemented during low-price



(a) Heat source of absorption chiller. (b) Heat supply for residential heating users.

Fig. 7. Comparison of heat utilization pathways in different seasons.

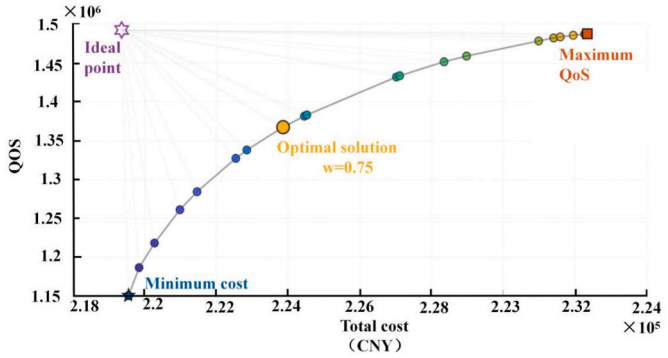


Fig. 8. Pareto frontier of total cost and quality of service (QoS).

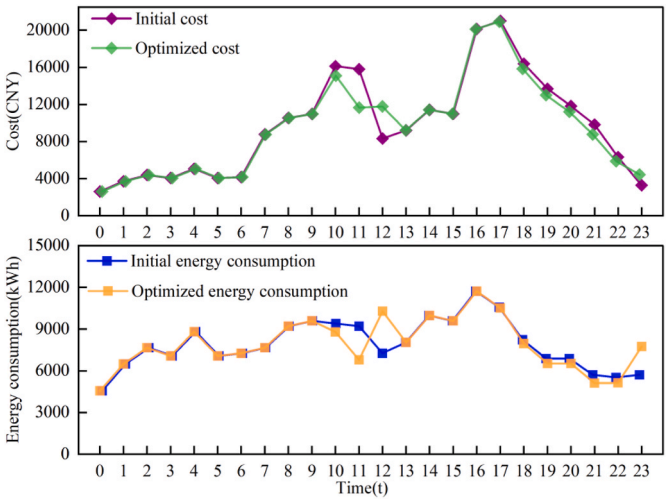


Fig. 9. Cost and energy consumption comparison before and after load shifting in the non-heating season.

periods.

Fig. 13 confirms that indoor temperatures remain within allowable limits in both heating and non-heating seasons. By effectively converting the server room into a short-term virtual thermal storage, air thermal inertia provides additional operational flexibility for cooling systems under time-varying electricity prices.

4.3. Comprehensive evaluation index analysis

This section conducts a comparative analysis of each scenario with the S1 baseline scenario based on the evaluation system established in Section 3.3.

As shown in Table 7, the introduction of the integrated energy system and flexible resources has improved system performance to varying degrees. Among them, scenarios S2 and S5 perform the best, with cost savings of 13% and 14.7%, respectively, and carbon emission reductions of 8.32% and 8.05%, respectively. The investment payback period for both scenarios is controlled within 5 years, indicating that the coordinated optimization of the integrated energy system and flexible scheduling can significantly enhance both the economic and environmental benefits of the data center.

The S3 scenario only considers load shifting, primarily guiding the shifting of some computing loads from high electricity price periods to low-price periods through time-of-use pricing to achieve economic operation. Since this strategy only optimizes the time distribution of energy usage without changing the overall energy consumption, the total carbon emissions remain unchanged, resulting in a carbon emission reduction rate of 0.

The S4 scenario introduces air thermal inertia, allowing the server room temperature to dynamically adjust within the 16–25°C range. Compared to the baseline scenario with a constant 20°C setting, this approach can slightly raise the room temperature during high electricity price periods and reduce cooling power, thereby slightly lowering energy consumption. However, due to the combined effects of factors such as diesel generator output and cooling tower heat dissipation, the overall carbon emissions slightly decrease, with the carbon emission reduction rate being slightly lower than the baseline value.

Overall, the results indicate that isolated flexibility strategies provide limited benefits, whereas their coordinated integration yields substantial economic and environmental improvements. By simultaneously leveraging integrated energy systems, computing load flexibility, and air thermal inertia, the proposed framework achieves significant cost reduction and carbon emission mitigation while maintaining acceptable investment payback periods. These findings provide practical insights for data center operators and planners seeking to balance economic performance, carbon reduction, and operational reliability.

4.4. Sensitivity analysis

Fig. 14 illustrates the impact of data center volume on the effectiveness of thermal inertia optimization. As the data center volume

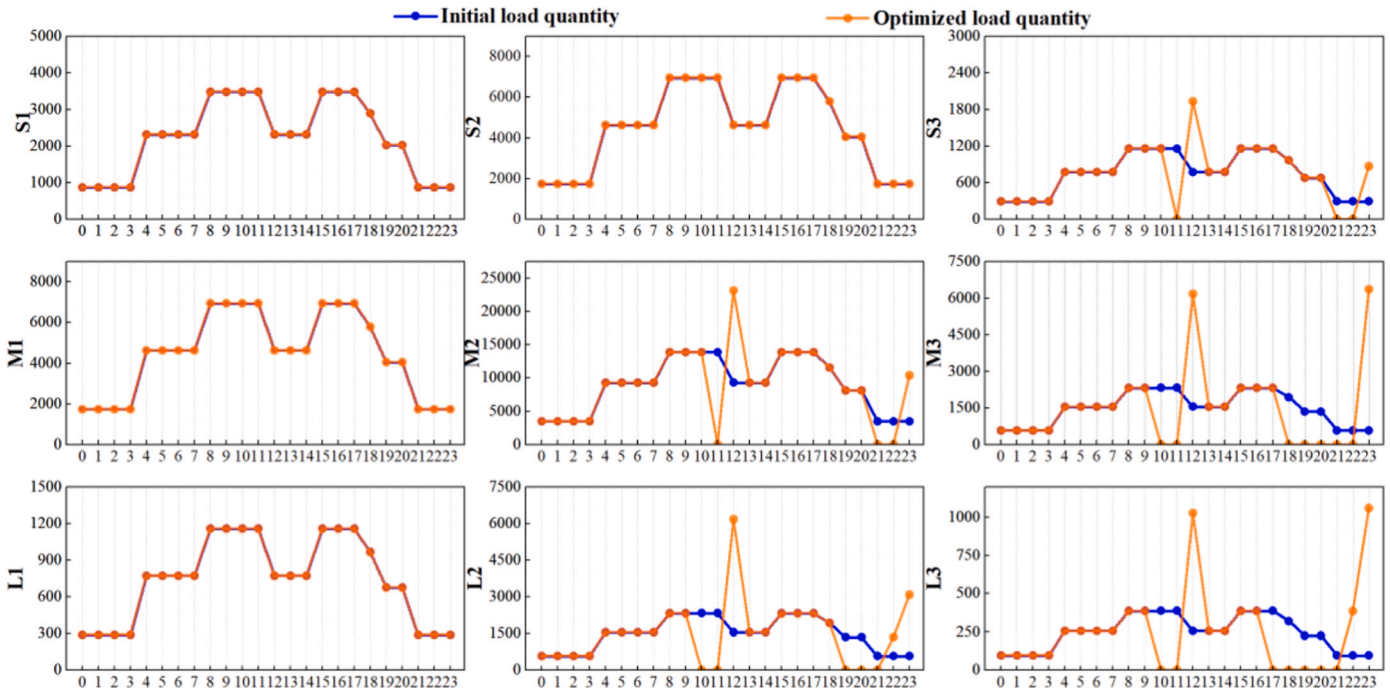


Fig. 10. Comparison of 9 types of batch load before and after optimization.

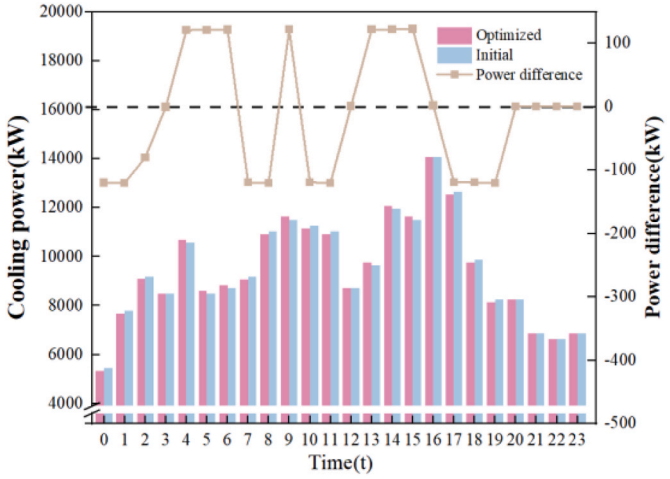


Fig. 11. Comparison before and after thermal inertia optimization in the non-heating period.

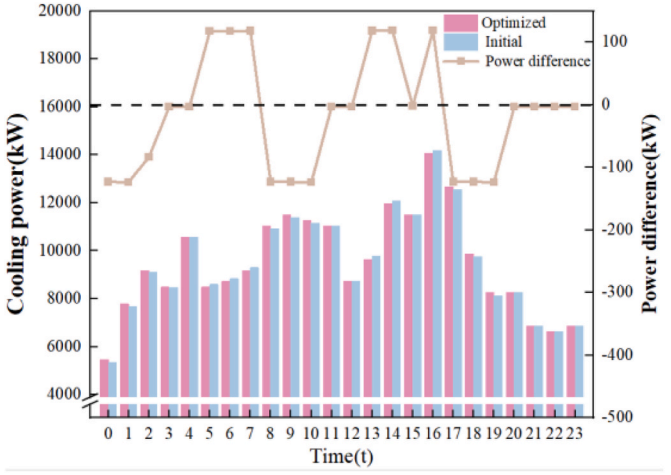


Fig. 12. Comparison of cooling power before and after thermal inertia optimization in the heating period.

increases, the total operating cost gradually decreases. This is because a larger volume increases the thermal capacity of the room air, thereby enhancing the system's thermal inertia and enabling the data center to store more cooling energy. Consequently, the load-shifting capability between cooling demand and electricity price periods is improved, reducing cooling demand during high-price periods. The results indicate that a larger data center volume is beneficial for utilizing thermal inertia regulation and improving the economic performance of the system.

Fig. 15 illustrates the variation of system operating cost with the batch-load ratio and allowable delay time. As the batch-load ratio increases from −30% to +30%, the operating cost decreases from 232,521 CNY to 224,484 CNY, indicating that a higher proportion of batch loads enhances load-shifting flexibility and reduces operating costs. Similarly, as the allowable delay time increases from $-3\Delta t$ to $3\Delta t$, the operating cost decreases from 239,474 CNY to 222,139 CNY. This demonstrates that extending the allowable delay time enlarges the scheduling flexibility and further improves the economic performance of the system.

5. Conclusion

This study proposes a coordinated optimization framework for data center integrated energy systems that explicitly integrates computational load flexibility and air thermal inertia to support the low-carbon transition of energy systems. By modeling the coupling relationships among electricity, cooling, heating, and computational subsystems, the proposed framework achieves coordinated optimization of energy conversion equipment and demand-side flexible resources under time-varying electricity prices and seasonal operating conditions.

A detailed case study based on a real data center in Hebei Province demonstrates the effectiveness of the proposed approach. The results show that the introduction of an integrated energy system fundamentally reshapes the optimal capacity configuration and operational strategy of the data center. In particular, waste heat recovery and absorption cooling significantly improve economic performance during the heating season, transforming waste heat from a passive byproduct

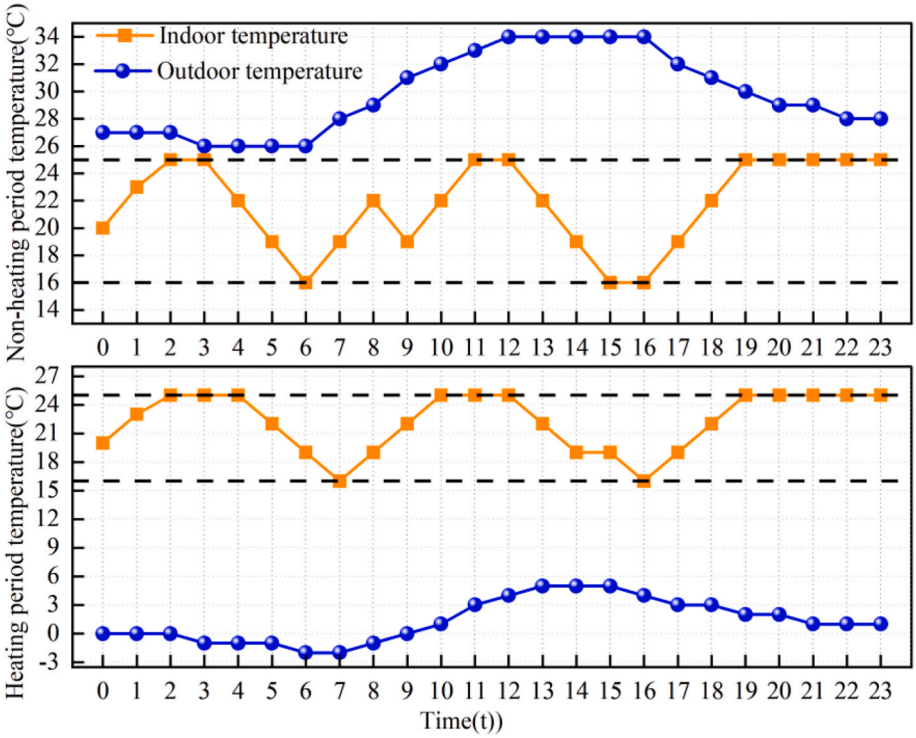


Fig. 13. Indoor and outdoor temperature curves of the data center in the non-heating and heating periods.

Table 7
Comparison of comprehensive evaluation indices.

	CSR(%)	ERR(%)	PBP(years)
S1	0	0	0
S2	13.48%	8.9%	4.55
S3	3.79%	0	0
S4	0.6%	0.93%	0
S5	15.13%	8.59%	4.08

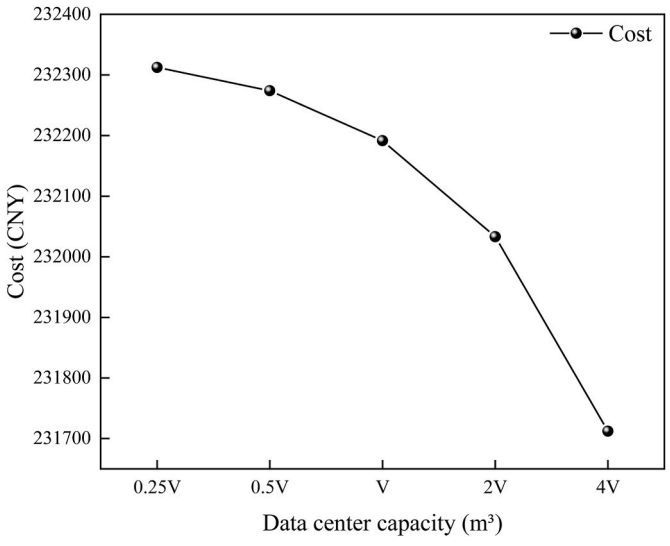


Fig. 14. Impact of data center volume on thermal inertia and cooling cost.

into an active energy resource.

Furthermore, the results indicate that isolated flexibility strategies, such as computing load shifting or air thermal inertia alone, provide limited economic and environmental benefits. In contrast, their

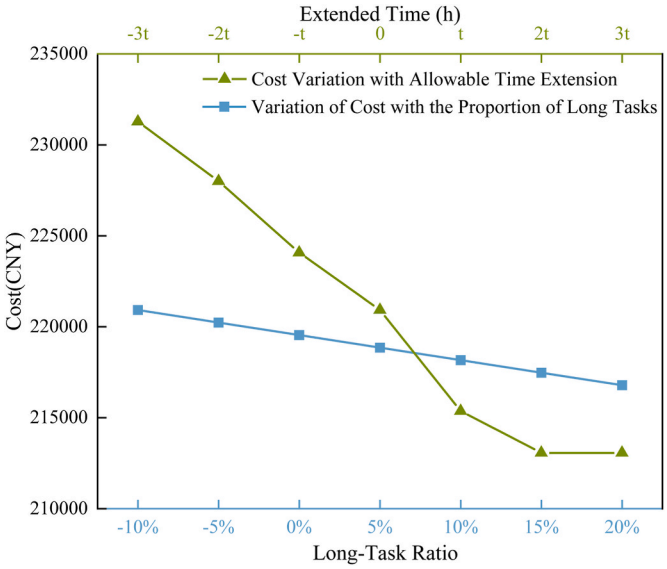


Fig. 15. Relationship between cost, batch-load ratio, and allowable time extension.

coordinated integration yields substantial synergistic effects. The combined strategy achieves up to 14.7% operating cost reduction and over 8% carbon emission reduction compared with the conventional operation mode, while maintaining acceptable service quality and keeping the investment payback period within five years. These findings highlight the importance of jointly exploiting supply-side and demand-side flexibility in data center energy system planning.

From an operational perspective, computing load shifting improves the temporal alignment between energy demand and electricity price signals without compromising quality of service, whereas air thermal inertia effectively converts the server room into a short-term virtual thermal storage, providing additional flexibility for cooling system

operation. Together with integrated energy systems, these flexibility dimensions enable data centers to actively participate in energy system optimization rather than behaving as rigid energy consumers.

Overall, this study provides practical insights for data center operators and planners seeking to balance economic performance, carbon reduction, and operational reliability. The proposed framework offers a scalable and shiftable solution for the low-carbon and cost-effective operation of data centers, particularly in regions characterized by pronounced electricity price fluctuations and heating demand. Future research may further extend the framework to multi-data-center coordination and incorporate uncertainty modeling to enhance its robustness under more complex operating conditions.

CRediT authorship contribution statement

Ligai Kang: Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Zhaoxun Gao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Conceptualization. **Lingzhe Zhang:** Resources, Project administration, Methodology. **Jixing Zhang:** Validation, Project administration, Conceptualization. **Qiaoyi Gao:** Software, Methodology, Funding acquisition. **Guohua Kong:** Conceptualization.

Generative AI statement

During the preparation of this manuscript, the authors used ChatGPT to improve language, clarity, and academic expression. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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